WEEK 4 · LESSON 21 OF 24 · HANDOUT

Privacy, Security and Safety

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=gEtTVlQsGcQ — open the lesson online to watch inside the course.

🎓 **Go deeper:** Andrej Karpathy: Intro to Large Language Models (1-hour talk) — https://www.youtube.com/watch?v=zjkBMFhNj_g

Your goal in this lesson: Explain the main privacy, security and safety risks of AI systems, describe key duties under Rwanda's data-protection law, and apply basic protections.

Three different questions

- **Privacy** — Is personal information collected and used fairly and lawfully?
- **Security** — Can attackers trick, steal or break the AI system?
- **Safety** — Could the system cause harm, even when nobody is attacking it?

Privacy and personal data

AI runs on data, and much of it is **personal**: names, phone numbers, ID numbers, locations, health records, photos and voice. Risks include:

- Collecting more data than needed.
- Using data for a purpose people never agreed to.
- Data **leaks** and breaches.
- Models that **memorise** and later repeat personal details from their training data.
- Users pasting confidential information into public chatbots.

Rwanda's data-protection law

Rwanda's **Law No. 058/2021 of 13/10/2021 relating to the protection of personal data and privacy** sets the rules. The **National Cyber Security Authority (NCSA)** supervises it. Key ideas every AI builder in Rwanda should know:

- Have a **lawful basis** (such as clear consent) for processing personal data.
- Collect only what you **need**, for a stated **purpose**.
- Keep data **accurate** and **secure**, and not longer than necessary.
- Respect people's **rights** — to be informed, to access their data, to correct it and to object.
- Organisations that process personal data must **register** with the supervisory authority.
- Personal data should generally be **stored in Rwanda**, unless the authority allows otherwise.

Always check the current official text and guidance from NCSA before building a real system.

Privacy protections

- **Data minimisation** — use only the data you need.

- **Anonymisation / pseudonymisation** — remove or replace names and IDs.
- **Access control** — only the right people can see the data.
- **Encryption** — protect data when stored and when sent.
- **Never paste** customer, patient or company secrets into public AI tools unless your organisation allows it.

Security threats to AI

Threat	What it is	Example
Adversarial examples	Tiny changes to input that fool a model	Stickers on a road sign confuse a car's camera
Data poisoning	Attackers add bad examples to training data	Fake reviews that teach a model wrong patterns
Prompt injection	Hidden instructions that hijack a chatbot or AI agent	A web page tells an AI assistant to leak the user's data
Model theft	Copying a model by querying it many times	A competitor clones a paid model
Deepfakes	Realistic fake audio, images or video	A fake voice of a manager asking staff to send money

Safety

Even without attackers, AI can cause harm: wrong medical advice, a self-driving error, a chatbot encouraging dangerous behaviour, or an automated system that fails silently. Safety practices include:

- **Testing** in realistic conditions, including rare and difficult cases.
- **Human oversight** for high-stakes decisions.
- **Clear limits** — tell users what the system can and cannot do.
- **Monitoring and a way to switch off** or roll back the system.
- **Reporting channels** so users can flag problems.

Researchers also study **alignment**: making sure powerful AI systems pursue the goals people actually intend.

Remember this: Protect personal data (follow Law No. 058/2021), defend AI systems against attacks like poisoning and prompt injection, and design for safety with testing, oversight and monitoring.

Name: _____ Date: _____

Activity

Task 1

Take your Week 2 project plan. List every piece of personal data it uses and whether you truly need it.

Task 2

Write a one-paragraph privacy notice a user would understand.

Task 3

Name one security threat and one safety risk for your system, with one protection for each.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which question is about privacy?

   A. Is personal information collected and used fairly and lawfully?

   B. Can attackers fool the model?

   C. How many layers does the model have?

   D. How fast is the GPU?

2. Which law protects personal data in Rwanda?

A. The EU traffic law
B. A school rule
C. There is no such law
D. Law No. 058/2021 relating to the protection of personal data and privacy

3. What is data minimisation?

A. Deleting the model
B. Making files smaller
C. Collecting and using only the data you need
D. Using cheaper computers

4. Hidden text on a web page tells an AI assistant to leak the user's data. This is…

A. Feature engineering
B. Prompt injection
C. Clustering
D. Gradient descent

5. Attackers add fake examples to a model's training data. This is…

A. Data poisoning
B. Recall
C. Tokenisation
D. Early stopping

6. Which is a good safety practice for high-stakes AI?

A. No testing, to save time
B. Hiding all limits from users
C. Removing reporting channels
D. Human oversight, monitoring and a way to switch the system off

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